Accept or abstain? A pre-specified test of reliability gates calibrated on held-out chemistry for molecular property and reaction-yield models

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Abstract

A molecular property or reaction-yield model is useful to a chemist only when it is right on the query in front of them, yet a model's uncertainty is rarely checked on chemistry it has not seen. We tested an accept-or-abstain protocol in which a gate, scoring each query by its similarity to the training set and by ensemble spread, returns a prediction only when it accepts the query. The gate's threshold is set by Learn-then-Test to a target precision on a calibration fold held out by chemistry rather than at random. On public data (23,401 logP values from SangsterLogP with two overlapping external sets, and the Buchwald–Hartwig and Suzuki–Miyaura high-throughput yield datasets), we evaluated eight held-out tests against four hypotheses defined in advance. Ranking queries by the gate's score lowered the error rate at 30% coverage by 0.16 to 0.35 on seven of the eight tests. At a target precision of 0.8, the calibration fold mattered most for yields: gates calibrated on a random fold accepted almost every reaction with an unseen additive at a precision of 0.59 to 0.75, whereas gates calibrated on held-out additives reached a pooled precision of 0.85 to 0.87 on three of the four tests, at lower coverage. At a target of 0.9, random-fold gates came within 0.05 of the target on three of the four tests. For the random forest, fusing similarity with spread did no better than spread alone, and conformal intervals calibrated on a random fold lost coverage under shift (down to 0.64). On a fluorinated-compound hold-out, the fused score ranked errors no better than chance at 30% coverage, and the held-out-calibrated gate missed its target. All four hypotheses therefore failed their success criteria. In an exploratory analysis, conformal intervals calibrated on the held-out fold recovered coverage on seven of the eight tests, at the cost of wider intervals. We release the protocol, the code and the complete results.

Keywords: applicability domain, selective prediction, conformal prediction, uncertainty quantification, lipophilicity, reaction yield

1 Introduction

Machine-learning models are now routine in medicinal chemistry and synthesis planning. They estimate lipophilicity, solubility and toxicity before a compound is made, and they score reaction conditions before a plate is run. A chemist uses such a prediction one compound or one reaction at a time. What matters to them is therefore not the average error over a benchmark, but whether this particular prediction can be trusted.

The usual way of reporting accuracy does not answer that question. Most models are evaluated on a random split of one dataset, so the test molecules resemble the training molecules closely. It is well documented that random splits overstate performance on new chemistry, and that scaffold, cluster and time splits give a more sober picture.[1–3] Our own earlier work met the same problem in three settings. A graph network

for chemical yields, comparable with the best published models on random splits, lost much of its accuracy on Buchwald–Hartwig reactions with unseen additives.[4] A binding-affinity graph network that reached an R^2 of 0.50 on random splits fell to 0.11 on the hardest of its held-out splits.[5] Published logP models, including the Crippen atom-contribution method[6] and pre-trained graph networks, were inapplicable to saturated fluorine-containing compounds, while models trained on the new measurements were not.[7]

The idea that a model should be used only inside its applicability domain is as old as the OECD principles for QSAR validation.[8] Regulators now ask a similar question of AI models in general: in its 2025 draft guidance the U.S. Food and Drug Administration frames model credibility in terms of a stated context of use.[9] In practice, however, few published molecular models state where they may be used, and fewer still check that statement on held-out chemistry.

A practical answer is to let the model abstain. If a model returns a prediction only when it is likely to be right, and says "do not know" otherwise, the precision of the predictions it does return can be controlled at the cost of coverage. Two kinds of signal are commonly used to decide: the distance of the query from the training data, and the model's own uncertainty. Both are established, and neither dominates in published benchmarks.[10,11] What is less often tested is how the decision threshold is chosen. A threshold calibrated on a random fold is calibrated on queries that look like the training set, and there is no reason to expect it to hold on chemistry that does not.

This work tested that question directly. We built an accept-or-abstain gate from a similarity signal and an ensemble-spread signal, set its threshold with a finite-sample procedure to a stated target precision, and compared calibration on a random fold with calibration on a fold held out by chemistry. We set four hypotheses, with success criteria, before the evaluation: that the gate lowers the error rate on held-out chemistry (H1); that random-fold calibration misses its target there while held-out-fold calibration meets it (H2); that combining the two signals ranks errors better than either alone (H3); and that gating restores the coverage of conformal intervals lost under shift (H4).

Hence, our contribution is threefold:

1. We describe a six-step accept-or-abstain protocol for molecular models and release a reference implementation under the MIT licence.

2. We evaluated it on public lipophilicity and reaction-yield data against eight held-out tests, including the four published out-of-sample additive tests of the Buchwald–Hartwig dataset and a fluorinated-compound hold-out built from our measured logP data.

3. We report the outcome of all four hypotheses, including those that failed, together with an operating-limits card for each held-out test.

1.1 Related works

The applicability-domain literature in QSAR is extensive. Sheridan et al. showed that the similarity of a query to its nearest training molecules is a good discriminator of prediction accuracy.[12] Tetko et al. compared distance-to-model measures on environmental-toxicity models and found that the spread of an ensemble of models was among the most reliable.[13] Kaneko and Funatsu likewise defined an applicability domain from ensemble disagreement.[14] Tropsha's review of best practices made the applicability domain a required part of model reporting.[15]

Conformal prediction gives intervals with a guaranteed marginal coverage when calibration and test data are exchangeable.[16,17] Norinder et al. introduced it to QSAR as an alternative to applicability-domain rules.[18] The guarantee does not survive a change in the distribution of inputs; Tibshirani et al. showed how

to repair it when the likelihood ratio of the shift is known,[19] which in chemistry it rarely is. Morger et al. showed on time-split bioactivity data and on toxicity endpoints that updating the calibration set with data more similar to the test set generally restores the validity of conformal predictors under data drift, at a cost in efficiency.[20]

Selective prediction treats abstention as part of the model. Geifman and El-Yaniv analysed the risk–coverage trade-off of selective classifiers.[21] Learn-then-Test by Angelopoulos et al. turns the choice of an abstention threshold into a multiple-testing problem, which yields a threshold whose risk is controlled with a stated probability for exchangeable data.[22] For molecular property prediction, Hirschfeld et al. and Scalia et al. benchmarked uncertainty estimates for graph networks, including deep ensembles,[23] and found that their ranking of errors depends on the dataset and on the split.[10,11]

On reaction yields, the Buchwald–Hartwig dataset of Ahneman et al. came with four out-of-sample tests on additives unseen in training.[24] Chuang and Keiser showed that much of the reported accuracy could be obtained from features without chemical meaning.[25] Schwaller et al. reported strong random-split results with a transformer and weaker results on the out-of-sample tests,[26] as did our directed message-passing network.[4] The Suzuki–Miyaura dataset of Perera et al. has been used mostly with random splits.[27]

1.2 Objective of the study

The primary objective of this work was to find out whether a reliability gate, calibrated in a way a practitioner can reproduce, keeps its promise on chemistry the model has not seen. For this purpose we deliberately used simple, widely available ingredients: fingerprints, a random forest, a small neural ensemble, nearest-neighbour similarity and a binomial test. If the gate works with these, it can be added to most existing models; if it does not, the reasons are easier to see.

2 Data

2.1 Lipophilicity

The logP pool is the SangsterLogP dataset of Cirino et al.,[28,29] which contains 23,520 curated measurements. After parsing with RDKit,[30] removal of duplicates by InChIKey, and removal of every compound that occurs in either external set, 23,401 compounds remained. Two external sets were held out completely. E1 is the prospective set distributed with SangsterLogP (327 compounds); 182 of them are the amides measured for our 2025 study.[7] E2 is the set of 183 amide logP values from that study, as released in its public repository. In E2, 140 compounds contain an sp^3 carbon–fluorine bond (SMARTS `[CX4]F`), and 43 are their non-fluorinated parents or methyl analogues.

2.2 Reaction yields

The Buchwald–Hartwig (BH) dataset of Ahneman et al. contains 3,955 Pd-catalysed C–N couplings of aryl halides with *p*-toluidine, varying the aryl halide, ligand, base and an isoxazole additive.[24] We used the four published out-of-sample tests, in which five of the 22 additives appear only in the test rows. The Suzuki–Miyaura (SM) dataset of Perera et al. contains 5,760 couplings varying both coupling partners, the ligand (eleven ligands and a ligand-free control), the base and the solvent; we used the UV-area yield.[27] Both datasets were taken from the public redistribution that accompanied our earlier yield study.[4] For SM, each component name was mapped to its structure using a table written for this study, with spectator fragments such as the internal standard omitted.

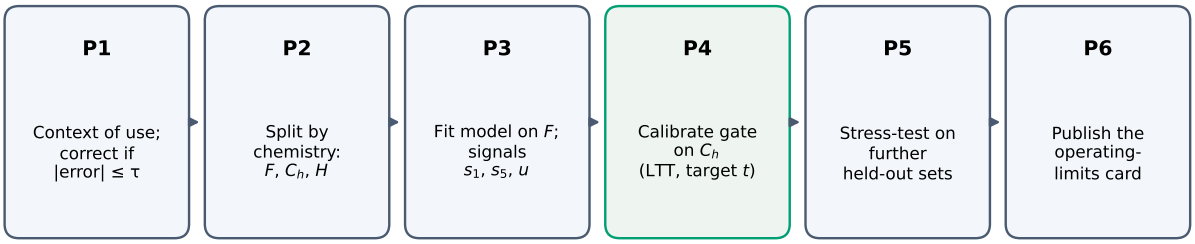


Figure 1. The accept-or-abstain protocol. P1: state the context of use and what counts as a correct prediction. P2: split the data by chemistry into a fit set F, a held-out calibration fold C_h and held-out tests. P3: fit the model and compute the signals. P4: calibrate the gate on C_h to a target precision. P5: test it on further held-out sets. P6: publish the operating limits.

It is worth mentioning that both reaction datasets come from single high-throughput campaigns. Their chemistry is narrow, and a large share of reactions gave little or no product. A "held-out" additive or ligand is therefore a small, controlled shift, and not new chemistry in the sense a medicinal chemist would use. We return to this point in Section 4.7.

3 Methodology

3.1 Protocol

The protocol has six steps (Figure 1). The user first states what a correct prediction is for the intended use (P1). In this study a logP prediction was correct if its absolute error was at most 0.5 log units, and a yield prediction if its absolute error was at most 10 percentage points. Coverage is the share of queries the gate accepts; selective precision is the share of accepted queries that are correct; the selective error rate is one minus selective precision. The data are then split by chemistry (P2), the model and its signals are fitted (P3), the gate is calibrated on a fold held out by chemistry to a target precision t (P4), it is tested on further held-out sets (P5), and the result is published as an operating-limits card (P6): for each target and test, the coverage and the precision actually achieved.

The hypotheses and their success criteria (Section 3.7) were set before the evaluation; the full analysis plan is given in the Supporting Information. Analyses outside the plan are marked as exploratory.

3.2 Representations and models

Molecules were represented by ECFP4 count vectors[31] (radius 2, 2,048 bins) together with ten RDKit descriptors (molecular weight, TPSA, numbers of H-bond donors and acceptors, rotatable bonds, rings, aromatic rings, heavy atoms and heteroatoms, and the fraction of sp^3 carbons). The Crippen logP descriptor was deliberately excluded, so that the model could not simply correct a known estimator. Reactions were represented by concatenating 1,024-bit ECFP4 vectors of each variable component.

The primary model (M1) was a random forest[32] of 300 trees with 20% of features considered at each split; its spread u is the standard deviation of the per-tree predictions. The secondary model (M2), used

Table 1. Datasets and splits. Sizes are means over five seeds. F: fit set; C_r: random calibration fold; C_h: held-out calibration fold; R: random (in-distribution) test; H: held-out test.

Task	Groups (number)	F	C_r	C_h	R	H	Held-out tests
LogP (D1)	Butina cluster (3,349)	11,429	2,449	3,522	2,449	3,551	cluster; E1 (327)
LogP, no sp^3 C–F (D1)	Butina cluster (3,160)	10,656	2,283	3,327	2,284	3,308	cluster; E2 (183)
Buchwald–Hartwig Test1–4 (D3a)	additive (17 + 5)	1,636	351	719	350	898	5 unseen additives
Suzuki–Miyaura (D3b)	ligand (12)	2,688	576	960	576	960	2 unseen ligands

only to check robustness, was an ensemble of five multilayer perceptrons (512 and 128 hidden units, ReLU, Adam, L2 penalty 10^{-4}, early stopping on 10% of the fit set); its spread is the standard deviation across members.[23] Both were implemented in scikit-learn.[33] Hyper-parameters were set in advance and not tuned.

3.3 Splits

For each task the pool was divided by a group label: Butina clusters[34] of ECFP4 at a Tanimoto distance of 0.65 for logP (3,349 clusters), the additive for BH, and the ligand for SM. Whole groups were assigned to a held-out test H, to a held-out calibration fold C_h, and to an in-distribution part. For logP, H and C_h each received about 15% of the compounds. For BH, H was the published test of each TestN, and four of the 17 training additives formed C_h. For SM, two ligands formed H and two others C_h; the ligand-free control counted as one of the twelve ligands and could be drawn for either role. The in-distribution part was split at random into the fit set F (70%), a random calibration fold C_r (15%) and a random test R (15%). Every model was trained on F alone, and both gates used the same model. Table 1 gives the sizes.

Holding out calibration additives leaves about 1,640 BH reactions in F, roughly half of the published training rows, so ungated BH accuracy is not directly comparable with published results on the same tests. To build the fluorinated hold-out, a separate logP model was trained after removing all 1,543 compounds with an sp^3 C–F bond from the pool, and was scored on E2. Five seeds (0–4) were used throughout.

3.4 Signals

Two signals of input novelty were computed against F: s_1, the largest Tanimoto similarity of the query fingerprint to any compound or reaction in F, and s_5, the mean of the five largest. The confidence signal was the ensemble spread u. For conformal intervals with nominal 90% coverage, we used normalised split-conformal prediction[17] with the score $|y - \hat{y}|/(u + \varepsilon)$, where ε is 0.1 times the median spread on the calibration fold.

3.5 Gates

Every calibration fold was split at random into halves C_a and C_b. Gate B, the fused gate, is a logistic regression of correctness on $(s_1, s_5, \log u)$, fitted on C_a; its score is the predicted probability that the prediction is correct. For comparison, the single-signal gates use s_1 or $-u$ directly as the score.

Thresholds were set on C_b by Learn-then-Test.[22] Candidate thresholds are the score values in C_b, tested in a fixed sequence from the strictest one that accepts at least 50 points to looser ones. A threshold that accepts n points of C_b, k of them correct, passes if

$$p = \Pr\left[\mathrm{Bin}(n, t) \geq k\right] \leq \delta, \qquad \delta = 0.1, \tag{1}$$

and the loosest threshold reached before the first failure is used. If no threshold passes, the gate abstains on every query. For queries exchangeable with C_b, the selective precision of the chosen threshold is then at least t with probability at least $1 - \delta$. Because the candidate thresholds are taken from the calibration scores themselves, a common simplification, the guarantee is approximate. It also says nothing about queries that are not exchangeable with C_b, which is why the choice between C_r and C_h matters. Gate A, a conjunctive baseline without a guarantee, accepts a query if $s_1 \geq \tau_1$ and $u \leq \tau_2$, with the pair chosen from a 20×20 grid of quantiles on the whole calibration fold as the largest coverage whose empirical precision is at least t. Targets were $t = 0.8$ and 0.9.

3.6 Metrics and hypotheses

For each test we report RMSE, MAE and R^2 without a gate; the risk–coverage curve of each score, that is, the selective error rate when the k highest-scoring queries are accepted; the area under that curve (AURC); the coverage and achieved precision of each gate; and the empirical coverage of the conformal intervals. Results are averaged over seeds. Bootstrap intervals resample test queries; for test sets that are identical in every seed (E1, E2 and BH Test1–Test4), the same resample is used for all seeds. The held-out tests are the logP cluster test, E1 and E2 (the D1 family), and BH Test1–Test4 and the SM ligand test (the D3 family). E1 and E2 share 182 compounds and are therefore not independent tests. The hypotheses, decided on M1, were:

H1 On every held-out test, the fused score's selective error rate at 30% coverage is below the ungated error rate, with a 95% bootstrap interval of the difference that excludes zero.

H2 Over the 16 (test $\times$ target) cells, Gate B calibrated on C_r misses its target by more than 0.05 in at least half; calibrated on C_h, it reaches at least $t - 0.05$ in at least three quarters.

H3 Gate B's AURC is at least 10% lower than that of the better single signal on at least two held-out tests in each family.

H4 Conformal intervals calibrated on C_r cover at least 0.88 of R, fall below 0.85 on at least one held-out test, and recover to at least 0.88 on the queries accepted by Gate B (C_h, $t = 0.8$) wherever they fell.

4 Results and discussion

4.1 The accuracy gap

Most held-out tests were harder than the random test of the same task (Figure 2, Table 2); the two external logP sets were the exceptions. For logP, the RMSE rose from 0.68 on R to 0.90 on held-out clusters, and the share of predictions outside 0.5 log units rose from 0.37 to 0.49. On E1 the RMSE was slightly lower than on R (0.65 against 0.68), although the error rate was slightly higher (0.38 against 0.37). For the BH additive tests the RMSE rose 1.5- to 2.8-fold, from 6.3–7.6 to 10.8–17.8 percentage points, with R^2 of 0.81, 0.84, 0.65 and 0.55 on Test1–Test4. For SM, holding out two ligands raised the RMSE from 12.4 to 20.1, and R^2 varied from -0.90 to 0.73 across seeds (mean 0.23), depending on which ligands were held out. The lowest value came from seed 0, in which the held-out pair was Xantphos and the ligand-free control.

The fluorinated hold-out behaved differently. The model trained without any sp^3 C–F compound had a lower error rate on E2 (0.36) than on its own random test (0.37), and a lower RMSE (0.61 against 0.70). On the fluorinated compounds its error rate (0.34) was lower than on their parents (0.40). This does not show

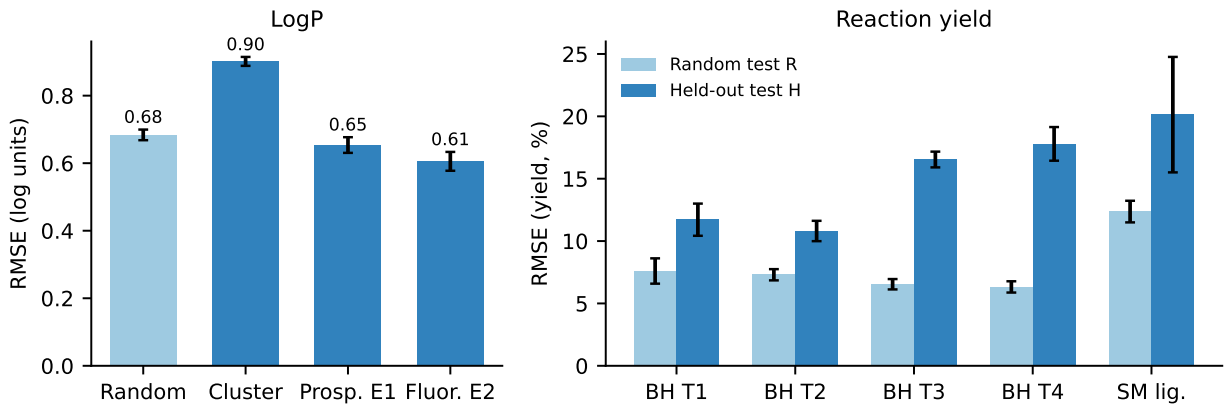


Figure 2. Ungated RMSE of M1 on the random test R and on the held-out tests (mean $\pm$ s.d. over five seeds). Left: logP. Right: reaction yields.

that the model had learned the effect of fluorination. Its R^2 on the sp^3 C–F compounds was -0.25, so it did not rank them, and the logP values in E2 span a narrow range: their standard deviation (0.59) is close to the 0.5 tolerance, and predicting the E2 mean for every compound would give an error rate of 0.40. The low error rate on E2 therefore reflects the narrow range of the set more than predictive skill, and the problem that motivated our earlier study of these compounds[7] is not solved by this model.

4.2 Which signal ranks errors

Ranking queries by the fused score lowered the error rate on seven of the eight held-out tests (Figure 3). At 30% coverage, the error rate fell from 0.49 to 0.29 on held-out logP clusters, from 0.38 to 0.22 on E1, from 0.27–0.41 to 0.04–0.20 on the four BH tests, and from 0.49 to 0.14 on the SM ligand test; every 95% interval excluded zero. On E2 it fell only from 0.355 to 0.345, with an interval of -0.098 to $+0.067$. Because H1 required every test, it does not hold, although it held on seven of eight.

The ranking came almost entirely from the ensemble spread. For M1, the similarity signal s_1 was weaker than $-u$ on every held-out test (Table 3): only slightly on E2 (AURC 0.337 against 0.323), markedly on the others. On the reaction tests it was far weaker than the spread, and on the SM ligand test it was worse than a random ordering. This is partly built into the design. For reactions, s_1 is computed on the concatenated fingerprints of all components, and a reaction with a new additive or ligand still shares its other components with many reactions in F, so its similarity barely registers the change. A component-wise similarity would be a fairer test of H3 for reactions. The fused score therefore learned to follow u, and its AURC was within 2% of that of $-u$ on every test, from 1.7% better to 0.9% worse. H3 does not hold for M1. This agrees with earlier reports that ensemble disagreement is one of the most reliable distance-to-model measures,[13,14] and with benchmarks in which no single uncertainty estimate dominated.[10,11]

In an exploratory check on E2, neither signal tracked the error: the rank correlation between absolute error and u ranged from -0.05 to 0.22 across seeds, compared with 0.28–0.38 on held-out clusters. The signals did register that the fluorinated compounds were unfamiliar: their mean s_1 was 0.43–0.46 across seeds, against 0.49–0.54 for the parents and 0.65–0.66 for the random test, and their spread was higher. But the model was not worse on them, so novelty and error were decoupled.

Table 2. Ungated accuracy of M1 (mean ± s.d. over five seeds). Error rate: share of predictions with an absolute error above 0.5 log units (logP) or 10 percentage points (yield).

Test set	n	RMSE	R^2	Error rate
LogP, random R	2,449	0.68 ± 0.02	0.84 ± 0.01	0.37
LogP, held-out cluster	3,551	0.90 ± 0.01	0.70 ± 0.02	0.49
LogP, prospective E1	327	0.65 ± 0.02	0.58 ± 0.03	0.38
LogP, fluorinated E2 (all)	183	0.61 ± 0.03	-0.07 ± 0.10	0.36
sp^3 C–F compounds	140	0.60 ± 0.03	-0.25 ± 0.13	0.34
parents	43	0.63 ± 0.02	0.23 ± 0.04	0.40
BH Test1, random R	350	7.60 ± 1.02	0.92 ± 0.02	0.14
BH Test1, held-out additives	898	11.71 ± 1.29	0.81 ± 0.04	0.28
BH Test2, random R	351	7.30 ± 0.45	0.93 ± 0.01	0.15
BH Test2, held-out additives	900	10.81 ± 0.82	0.84 ± 0.02	0.27
BH Test3, random R	350	6.54 ± 0.42	0.94 ± 0.01	0.12
BH Test3, held-out additives	897	16.54 ± 0.63	0.65 ± 0.03	0.37
BH Test4, random R	350	6.32 ± 0.45	0.95 ± 0.01	0.09
BH Test4, held-out additives	900	17.79 ± 1.35	0.55 ± 0.07	0.41
SM, random R	576	12.36 ± 0.87	0.81 ± 0.03	0.32
SM, held-out ligands	960	20.13 ± 4.63	0.23 ± 0.65	0.49

Table 3. AURC of the selective error rate for M1 (lower is better; mean over seeds). "No gate" is the ungated error rate, i.e., the AURC of a random ordering.

Held-out test	No gate	s_1	$-u$	Fused
LogP cluster	0.494	0.436	0.343	0.342
LogP prospective (E1)	0.385	0.317	0.267	0.263
LogP fluorinated (E2)	0.355	0.337	0.323	0.322
BH Test1	0.281	0.259	0.112	0.112
BH Test2	0.270	0.267	0.097	0.097
BH Test3	0.365	0.322	0.164	0.164
BH Test4	0.412	0.337	0.251	0.251
SM ligand	0.489	0.503	0.271	0.273

4.3 The calibration split

Where the calibration fold came from mattered most for reaction yields (Figure 4, Table 4). On the BH additive tests, gates calibrated on C_r accepted 94–99% of reactions and reached a precision of only 0.59–0.75 against a target of 0.8. A random calibration fold contains reactions with additives seen in training; on those, the model is accurate and the threshold is loose. Gates calibrated on C_h, whose additives were also unseen, reached a pooled precision of 0.87, 0.85 and 0.85 on Test1–Test3 (seed-averaged 0.88, 0.87 and 0.90). They did so by accepting fewer reactions: 44%, 62% and 29%. Test4 was the exception. There the C_h gate reached only 0.64, at 70% coverage. Test4 is also the hardest test without a gate (error rate 0.41), and its additives evidently shifted yields more than the four calibration additives did. On the SM ligand test the two folds gave similar pooled precision (0.73 for C_r, 0.74 for C_h), and the C_h gate accepted queries in only two of the five seeds. At the stricter target of 0.9, the difference between the folds was smaller: C_r-calibrated gates reached a pooled 0.91, 0.87 and 0.85 on BH Test1–Test3, within 0.05 of the target, and only Test4 (0.65) fell well short (Table S4).

For logP, both calibration folds gave similar precision on the held-out tests, and both gates were very

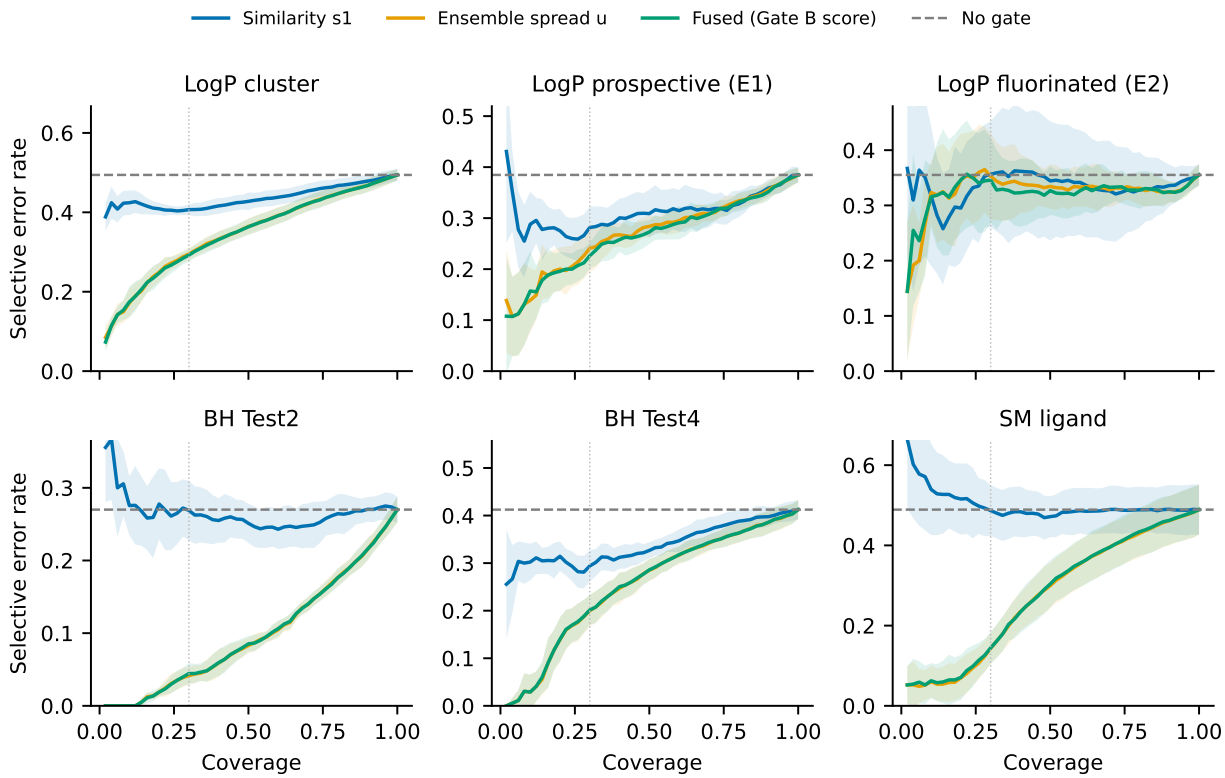


Figure 3. Risk–coverage curves of M1 on six held-out tests (mean ± s.d. over seeds): selective error rate when the queries with the highest score are accepted. The fused score is fitted on C_h. The dashed line is the ungated error rate; the dotted line marks 30% coverage (H1). BH Test1 and Test3, not shown, behave like Test2 (Table 3).

conservative. On held-out clusters the C_h gate accepted 5% of compounds at a pooled precision of 0.82, and it abstained completely in two of the five seeds. A random fold was not over-optimistic here, possibly because the threshold that reaches the target is so strict that it selects compounds close to the training set in either case; we did not test this.

Counting cells, C_r-calibrated gates missed the target by more than 0.05 in 6 of the 16 cells, short of the 8 required, while C_h-calibrated gates met $t - 0.05$ in 13 of 16, above the 12 required. H2 therefore does not hold as a whole: its second part holds and its first does not. The misses of the C_r gate were all on reactions, apart from the 0.8 target on E2. The second part is borderline. Several of its cells rest on a single accepting seed with few compounds (E2 at $t = 0.9$: two compounds; E1 at $t = 0.9$: ten), and with pooled instead of seed-averaged precision it falls to 12 of 16, exactly at the threshold.

Two properties of the gate explain much of this. First, the test is strict at the start of the sequence. With 50 accepted calibration points, it needs 45 correct to certify $t = 0.8$ and 49 to certify $t = 0.9$; when the 50 highest-scoring calibration points fall short of that, the gate abstains on everything, which is what happened in the seeds that abstained. Second, the guarantee is weaker in practice than in principle. In an exploratory count over single seeds, the C_h-calibrated gate met its target in 31 of the 45 seed-cells in which it accepted anything (69%), and the C_r-calibrated gate in 15 of 67 (22%), whereas Learn-then-Test promises at least 90% for exchangeable data. Held-out calibration brings the gate much closer to its promise, but not all the way.

The conjunctive Gate A, which has no guarantee, gave similar precision at somewhat higher coverage. On BH Test1, for example, both reached a seed-averaged precision of 0.88, Gate A at 57% coverage and

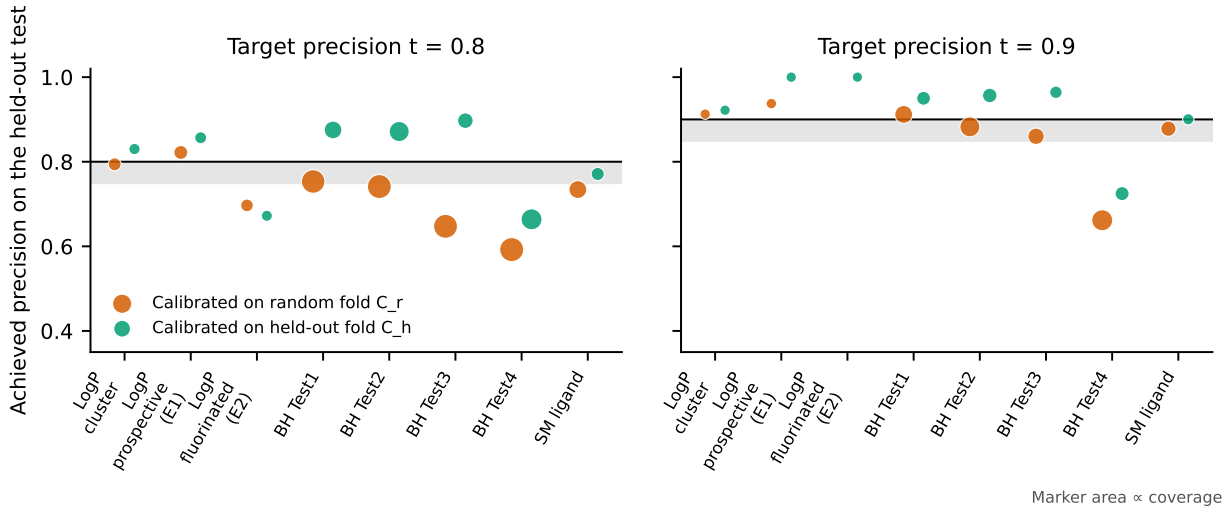


Figure 4. Precision achieved by Gate B (M1) on each held-out test when its threshold is calibrated on the random fold C_r or on the held-out fold C_h (mean over seeds that accepted at least one query). The black line is the target; the grey band is the 0.05 tolerance of H2. Marker area is proportional to coverage.

Gate B at 44%; on the SM ligand test Gate A accepted 26% of reactions at 0.85, against 14% at 0.77 for Gate B. The difference reflects the cost of the $1 - \delta$ guarantee and of halving the calibration fold.

4.4 Conformal coverage under shift

Calibrated on C_r, the 90% intervals covered 0.88–0.92 of every random test, as expected (Figure 5). Under shift they did not: coverage fell to 0.846 on held-out logP clusters, 0.83 and 0.82 on BH Test1 and Test2, 0.76 on Test3, 0.64 on Test4 and 0.76 on the SM ligand test. On logP clusters, the gate restored coverage to 0.91 on the accepted compounds. On the reaction tests it did not (0.62–0.85). A plausible reason is that the accepted reactions still carry the systematic offset of an unseen additive or ligand, which the ensemble spread does not see. H4 does not hold.

In an exploratory analysis we calibrated the same intervals on C_h instead. They then covered at least 0.90 of the queries on seven of the eight held-out tests (0.90–0.96), and 0.80 on BH Test4, in line with the finding of Morger et al. that a calibration set closer to the test data generally restores validity at a cost in efficiency.[20] The price is width: the intervals were about 1.2 times as wide as the C_r-calibrated ones for logP and 1.6–2.2 times for yields. On the SM ligand test they were 86 percentage points wide on average, too wide to be useful.

4.5 The fluorinated case

Figure 6 shows the fluorinated case in detail. The gate declined almost every fluorinated compound: it accepted 0.6% of them, against 14% of the parents. Ungated precision on the fluorinated compounds was 0.66, already below the target of 0.8, so declining them was consistent with the target. But the compounds the gate did accept, mostly parents, were predicted with a precision of only 0.67. The signals registered novelty (the fluorinated compounds had lower similarity and higher spread) without separating right from wrong predictions. Simpler gates did somewhat better on E2: the spread-only gate and Gate A reached a seed-averaged precision of 0.77 and 0.78, at 4% and 10% coverage.

Table 4. Operating-limits card for Gate B (M1) at target precision $t = 0.8$. Coverage is the mean over five seeds, including seeds that abstained on every query. Precision is pooled over all accepted queries of all seeds, with a Wilson 95% interval. "Seeds accepting" counts the seeds in which the C_h-calibrated gate accepted at least one query. H2 is judged on seed-averaged precision (Figure 4).

	Calibrated on C_r		Calibrated on C_h		
Held-out test	Coverage	Precision [95% CI]	Coverage	Precision [95% CI]	Seeds accepting
LogP cluster	0.13	0.79 [0.78, 0.81]	0.05	0.82 [0.79, 0.84]	3/5
LogP prospective (E1)	0.19	0.78 [0.73, 0.82]	0.08	0.83 [0.75, 0.88]	3/5
LogP fluorinated (E2)	0.12	0.72 [0.62, 0.79]	0.04	0.71 [0.54, 0.83]	3/5
BH Test1	0.94	0.75 [0.74, 0.77]	0.44	0.87 [0.86, 0.89]	4/5
BH Test2	0.98	0.74 [0.73, 0.75]	0.62	0.85 [0.84, 0.86]	5/5
BH Test3	0.98	0.65 [0.63, 0.66]	0.29	0.85 [0.83, 0.86]	4/5
BH Test4	0.99	0.59 [0.58, 0.61]	0.70	0.64 [0.62, 0.65]	5/5
SM ligand	0.45	0.73 [0.71, 0.75]	0.14	0.74 [0.70, 0.77]	2/5

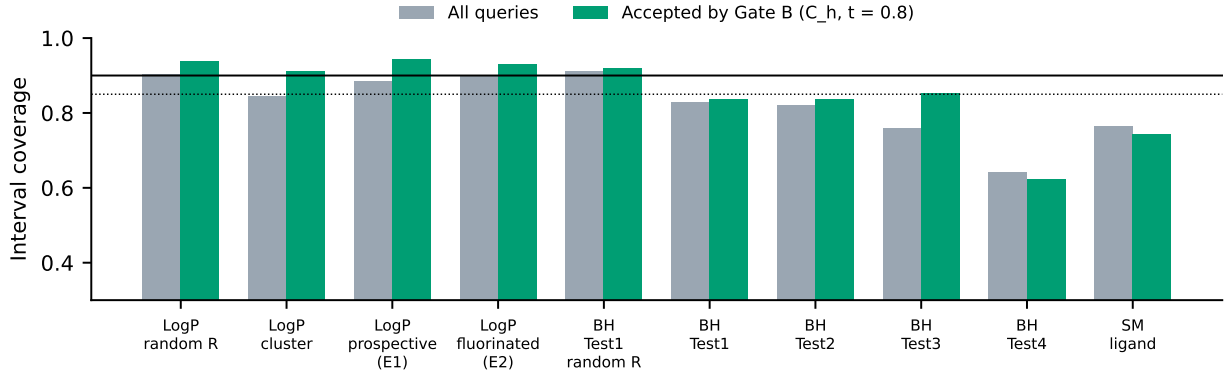


Figure 5. Empirical coverage of normalised conformal intervals (nominal 90%, calibrated on C_r) for M1, on all queries and on the queries accepted by Gate B (C_h, $t = 0.8$). Solid line: nominal level; dotted line: 0.85.

4.6 A second model family

The MLP ensemble (M2) was somewhat more accurate than the random forest on random and held-out logP clusters (RMSE 0.62 and 0.84, against 0.68 and 0.90) and on the random BH tests, and similar on held-out additives (Table S1, Supporting Information). It generalised worse to the fluorinated hold-out: its error rate on E2 was 0.48, against 0.36 for M1, and 0.50 on the sp^3 C–F compounds alone. Its signals also behaved differently. On the logP tests, the similarity signal ranked errors better than the spread of the MLP ensemble, and the fused score improved on the better single signal by 7.4%, 2.8% and 5.7% (Table S2). That is more than for M1, but short of the 10% that H3 requires. On reactions, the fused score was no better than spread, and on the SM test it was 7% worse. H3 therefore fails for M2 too.

H1 held for M2 on all eight tests, although the reductions on reactions were small (0.03 to 0.09). The C_h-calibrated gate of M2, however, abstained on every query of every held-out test (Table S3): on the held-out calibration folds, no threshold that accepted at least 50 points passed the binomial test (Section 4.3). On the BH Test1 calibration folds, for example, the 50 highest-scoring reactions had a precision of only 0.58–0.74. H2 therefore holds for M2 only vacuously, because a gate that accepts nothing counts as not missing its target, and H4 cannot be checked. The C_r-calibrated gate of M2 missed its target on every BH

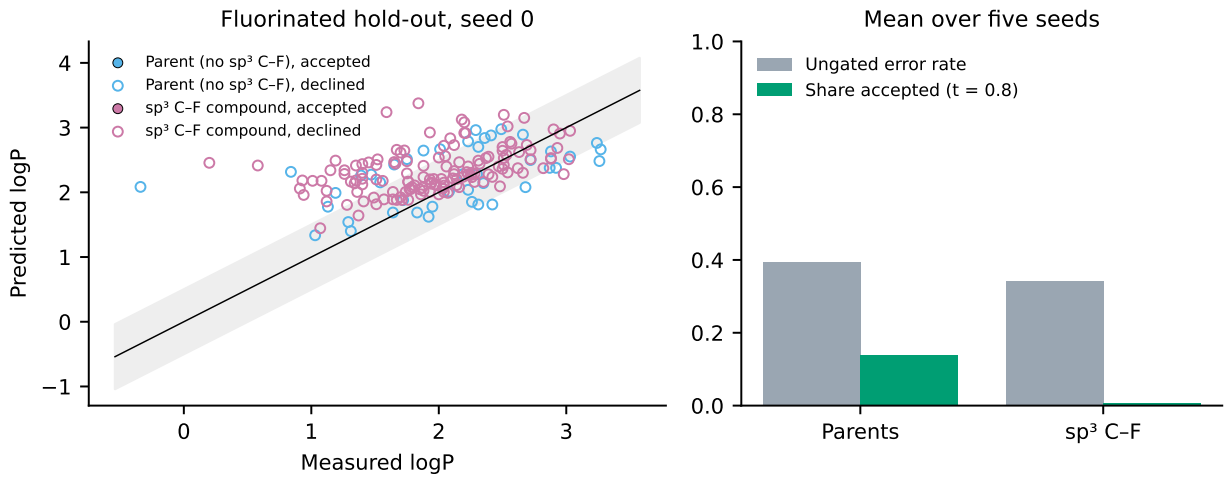


Figure 6. The fluorinated hold-out (M1 trained without sp^3 C–F compounds). Left: predicted against measured logP for E2 in one seed; filled markers were accepted by Gate B (C_h, $t = 0.8$); the grey band is ± 0.5 log units. Right: ungated error rate and share accepted for parents and for sp^3 C–F compounds, averaged over seeds.

Table 5. Hypotheses and their outcomes. M1 decides; M2 is a robustness check. For M2, H2 holds only vacuously: its C_h-calibrated gate abstained on every query of every held-out test, which counts as not missing the target.

Hypothesis	M1 (RF)	M2 (MLP)
H1: gate lowers error at 30% coverage	does not hold	holds
H2: calibration split matters	does not hold	holds (vacuously)
H3: fusion beats best single signal	does not hold	does not hold
H4: gating restores conformal coverage	does not hold	does not hold

test by a wider margin than M1, with a precision of 0.52–0.70 at $t = 0.8$. Table 5 summarises the outcomes.

4.7 Limitations

The study is retrospective: every split was drawn from existing data, and no prediction was tested by new measurement. The reaction datasets come from two high-throughput campaigns with narrow chemistry, so a held-out additive or ligand is a modest shift. Their results say little about yields on diverse substrates. E2 is small (183 compounds), and its narrow logP range makes R^2 uninformative. We used fingerprint models and a small neural ensemble; graph networks and other uncertainty estimates may rank errors differently. The gate's threshold test starts at 50 calibration points, which makes it abstain whenever the most confident queries are not almost all correct; other starting points or a Bonferroni correction over the grid would behave differently. The SM test touched only seven of the twelve ligands across five seeds, and per-ligand results vary widely. The correctness thresholds of 0.5 log units and 10 percentage points are choices for illustration; a real context of use would set its own. Finally, a shift that leaves the inputs looking familiar while changing the outcome, such as a change of assay protocol, is invisible to any input-based signal.

5 Conclusions

A reliability gate is easy to build from signals most models already have, and on seven of eight held-out tests ranking queries by such a gate lowered the error rate substantially. The way the gate's threshold is calibrated mattered more than the choice of signal. At a target precision of 0.8, a threshold calibrated on a random fold, as is usual, accepted nearly all reactions with an unseen additive at a precision well below its target. A threshold calibrated on a fold held out in the same way as the intended use came much closer to the target, at lower coverage. On the other hand, for the random forest, combining similarity with ensemble spread added nothing to spread alone. Conformal intervals calibrated on a random fold lost coverage under shift, and gating did not restore it for reactions; calibrating them on a held-out fold did, in an exploratory analysis, at the cost of width. On the fluorinated hold-out the signals registered novelty but did not separate right from wrong predictions. For the primary model, all four hypotheses failed their criteria. A neural ensemble ranked errors on every held-out test, but its held-out-calibrated gate could certify none of them. The practical recommendation is therefore narrower: calibrate abstention thresholds on data held out by chemistry, and publish the coverage and precision achieved on held-out tests, not on random ones.

Supporting Information

Tables S1–S3 (results of the MLP ensemble M2), Table S4 (operating-limits card of M1 at $t = 0.9$) and the analysis plan (PDF).

Data and code availability

All data are public (SangsterLogP, CC BY 4.0;[29] the Buchwald–Hartwig, Suzuki–Miyaura and fluorinated-amide data under the MIT licence of their repositories). The code, the analysis plan, the split definitions, all per-query predictions and the scripts that regenerate every figure and table are available at `https://github.com/blackthorn-ai/accept-or-abstain` under the MIT licence.

Author contributions

O.G.: conceptualization, methodology, supervision, validation, writing (review and editing).

Competing interests

O.G. is the director and owner of Blackthorn AI Ltd., which provides machine-learning services to life-sciences companies, including work on model reliability. No client data, code or results were used in this work.

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